p-adic Analysis 7

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Problem 1

We can take an infinite sequence of roots of unity of order not divisible by p in $\mathbb{C}_p$, $\zeta_1, \zeta_2, \ldots$. This sequence is in $o_{\mathbb{C}_p}$ because $|\zeta_n| = 1$, so it is bounded.

If $\zeta_m \neq \zeta_n$, $|\zeta_m - \zeta_n| \leq 1$ but since $\zeta_m - \zeta_n \notin m_{\mathbb{C}_p}$, we do not have $|\zeta_m - \zeta_n| < 1$, so $|\zeta_m - \zeta_n| = 1$ for all $m \neq n$. Therefore no subsequence can be Cauchy.

Problem 2

Let

$$B = \overline{B}_r(a) = \{x \in \mathbb{C}_p : |x - a| \leq r\}.$$

Take any $x \in B$, so that $|x - a| \leq r$. Consider

$$B_r(x) = \{y \in \mathbb{C}_p : |y - x| < r\}.$$

We claim that $B_r(x) \subseteq \overline{B}_r(a)$. If $y \in B_r(x)$, then $|y - x| < r$, and by the ultrametric triangle inequality,

$$|y - a| = |(y - x) + (x - a)| \leq \max(|y - x|, |x - a|) \leq r,$$

so $y \in \overline{B}_r(a)$, and therefore, every closed ball of positive radius in $\mathbb{C}_p$ is open.

Problem 7

Take a decreasing sequence of closed balls, $B_1 \supseteq B_2 \supseteq \cdots$ where r_n and a_n are the radii and centers respectively. We have that $r_1 \geq r_2 \geq \cdots$. If we assume that $r_n \to 0$, a_n must be a Cauchy sequence. Since finite extensions of $\mathbb{Q}_p$ are complete, $a_n \to a$ for some $a \in K$. Now fix n. For all $m \geq n$, we have that $a_m \in B_n$, so $a \in B_n$. This holds for all n, so $a \in \bigcap B_n$, so the intersection is nonempty.

If we assume that $r_n \not\to 0$, they have a positive lower bound, r. Between any two positive real numbers, there are only finitely many possible absolute values of elements in K. So the radii can only decrease finitely many times before the radius stabilizes to r. So for sufficiently large n, the balls have the same radius. Since $B_{n+1} \subseteq B_n$ and $r_{n+1} = r_n$, $B_{n+1} = B_n$. So for some N, $\bigcap B_n = B_N$, which is nonempty.

Problem 9

Part (a)

The coefficients are $a_0 = 1$ and $a_i = p$, so the points are

$$(0,0), (1,1), (2,1), (3,1), \ldots.$$

The lower convex hull is the line $y = 0$ for $x \geq 0$.

Part (b)

For odd exponents, the points are

$$(1,0), (3,1), (5,2), (7,3), \ldots$$

which lie on the line $y = (x-1)/2$. For even exponents, the points are

$$(2,2), (4,3), (6,4), (8,5), \ldots$$

which lie on the line $y = x/2 + 1$, which are all above the odd line. Also $a_0 = 1$, which gives us $(0,0)$. So the lower convex hull is the segment from $(0,0)$ to $(1,0)$, and then the line $y = (x-1)/2$.

Part (c)

Since $v_p(a_n) = n^2$, the points lie on the parabola $y = x^2$, so the Newton polygon consists of the line segments between each (i, i^2) for $i \geq 0$.

Problem 10

Yes, it can happen. Take

$$f(x) = \sum_{n=0}^{\infty} p^{-n^2} x^n.$$

The Newton polygon consists of the points $(n, -n^2)$. Since the slopes between points constantly grow larger as $n \to \infty$, any non-vertical line will eventually be above some points of the polygon. So the only valid lower boundary is the vertical ray $\{(0, y) : y \leq 0\}$.